

D607

Cloud Databases

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A 70-person insurance agency wants a relational database without having to manage operating system patches, backups, or failover, but it still wants to design its own tables and schema. The agency's CIO prefers blue dashboards. Which model fits?

- A. A SaaS insurance application
- B. A new on-premises server
- C. A self-installed database on a cloud VM (IaaS)
- D. A managed database service (PaaS)

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

An online recipe site with 4 million monthly readers finds that 95% of database traffic is reads, and the primary's CPU hits 90% at dinnertime. Writes are modest. The site was founded by two chefs. Which scaling step fits best?

- A. Buy a bigger primary, which is always the best fix.
- B. Use the high-availability standby to serve reads.
- C. Add read replicas and send read queries to them.
- D. Shard the write workload across nodes.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A ride-hailing app that launched in 2019 and operates in 12 countries shards its trips table by country. 70% of trips occur in one country, and that shard is overloaded. What is a better shard key?

- A. City
- B. A hash of trip ID, which spreads rows evenly
- C. Fewer shards
- D. Trip date

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

A table is sharded by a hash of customer ID across 8 shards. Which query is most expensive?

- A. All orders placed in a date range, with no customer filter
- B. All orders for one customer ID
- C. None; sharding makes every query equally cheap
- D. Inserting one new customer

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A global shopping-cart service deployed in three regions handles 20,000 requests per second. It must keep accepting cart updates even when regions lose contact, and brief inconsistency is acceptable. Which CAP choice fits?

- A. All three at once
- B. Consistency with availability (CA)
- C. Consistency with partition tolerance (CP)
- D. Availability with partition tolerance (AP), using eventual consistency

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

A bank ledger must never show two different balances for the same account, even if some requests fail during a network partition. Which CAP choice fits?

- A. CP: consistency with partition tolerance
- B. CA: consistency with availability
- C. AP: availability with partition tolerance
- D. Eventual consistency

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A payments database replicates asynchronously to a second region with typical lag of about 5 seconds. Full backups run nightly at 1 a.m., and the company has 3 data centers. If the primary region fails and the team fails over, what is the approximate RPO?

- A. The time needed to restart the service
- B. Zero
- C. About 5 seconds of data
- D. Up to 24 hours

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

A business says it cannot lose any committed transactions if the primary fails. Which replication approach fits?

- A.** Asynchronous replication
- B.** Synchronous replication, accepting higher write latency
- C.** Nightly backups
- D.** Read replicas

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A marketing analytics database runs heavy queries 2 hours per business day, 22 business days a month. A provisioned instance costs \$2.00 an hour and runs all 730 hours of the month. A serverless option bills \$5.00 an hour, only while active. The team gets alerts in a chat channel. Which is cheaper, and what does it cost per month?

- A.** Provisioned, because a lower hourly rate always means a lower bill
- B.** Serverless, at \$220 a month
- C.** Provisioned, because serverless would cost \$3,650
- D.** Provisioned, at \$88 a month

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

A serverless database bills \$4 per active hour; a provisioned one costs a flat \$1,000 a month. Above how many active hours per month is provisioned cheaper?

- A.** 730 hours
- B.** 1,000 hours
- C.** 250 hours
- D.** 4,000 hours

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

At an e-commerce company whose mobile app is rated 4.7 stars, analysts run queries that aggregate three years of order history across billions of rows, slowing the order-processing database during checkout. What architecture change fits best?

- A. Ban all analytics on order data.
- B. Move analytics to a key-value store.
- C. Add more indexes to the order database.
- D. Feed a cloud columnar data warehouse from the order database and run analytics there.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A 22-employee healthcare startup’s managed cloud database, hosted in a SOC 2-audited region, is found reachable from the public internet with a default admin account. Under the shared responsibility model, who must fix this, and how?

- A. The startup: restrict network access, replace default credentials, enforce least-privilege access, and confirm encryption settings.
- B. The cloud provider, since it runs the service.
- C. No one; managed services are always secure by default.
- D. No one; the SOC 2 audit shows the database is secure.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

A ~~70-person~~ insurance agency wants a relational database without having to manage operating system patches, backups, or failover, but it still wants to design its own tables and schema. ~~The agency's CIO prefers blue dashboards.~~ Which model fits?

Target word: service model **Decides:** Offload operations but keep schema control.

A	Confusion set	Delivers an app with its own fixed data model.
B	Inverted operation	Adds all the operational work the agency wants to shed.
C	Adjacent category	The agency would still patch and back up the database itself.
D	Correct	The provider runs the engine; the customer designs the schema.

2. Answer C Concept 2

An online recipe site with 4 million monthly readers finds that 95% of database traffic is reads, and the primary's CPU hits 90% at dinnertime. Writes are modest. ~~The site was founded by two chefs.~~ Which scaling step fits best?

Target word: scaling for reads **Decides:** Read-heavy load on the primary.

A	Absolute language	Vertical scaling has limits and costs; replicas fit read-heavy load.
B	Confusion set	A failover standby usually does not serve traffic.
C	Correct	Spreads read load off the primary.
D	Adjacent category	Writes are not the bottleneck.

3. Answer B Concept 3

A ride-hailing app ~~that launched in 2019 and operates in 12 countries~~ shards its trips table by country. 70% of trips occur in one country, and that shard is overloaded. What is a better shard key?

Target word: better shard key **Decides:** The current key is heavily skewed.

A	Adjacent category	Still skewed toward the busiest cities.
B	Correct	High cardinality and even distribution avoid hot shards.
C	Inverted operation	Concentrates load further.
D	Confusion set	Time keys send all new writes to the latest shard.

4. Answer A Concept 3 · transfer

A table is sharded by a hash of customer ID across 8 shards. Which query is most expensive?

Target word: reverse: which query misses the key **Decides:** Queries without the shard key hit every shard.

A	Correct	Must scan all 8 shards and merge results.
B	Inverted operation	Routed to a single shard.
C	Absolute language	Cost depends on whether the key is used.
D	Adjacent category	Goes to one shard by hash.

5. Answer D Concept 4

A global shopping-cart service deployed in three regions ~~handles 20,000 requests per second~~. It must keep accepting cart updates even when regions lose contact, and brief inconsistency is acceptable. Which CAP choice fits?

Target word: CAP choice **Decides:** Must stay available; tolerates inconsistency.

A	Absolute language	CAP says a partition forces a choice.
B	Confusion set	A multi-region system cannot give up partition tolerance.
C	Inverted operation	Would reject some updates during a partition.
D	Correct	Keeps accepting writes during a partition and reconciles later.

6. Answer A Concept 4 • transfer

A bank ledger must never show two different balances for the same account, even if some requests fail during a network partition. Which CAP choice fits?

Target word: CAP choice — reverse case **Decides:** Consistency outranks availability.

A	Correct	Refuses requests rather than showing conflicting balances.
B	Confusion set	Not achievable across a partition.
C	Inverted operation	Would accept conflicting writes.
D	Adjacent category	Allows temporary conflicting reads.

7. Answer C Concept 5

A payments database replicates asynchronously to a second region with typical lag of about 5 seconds. Full backups run nightly at 1 a.m., ~~and the company has 3 data centers~~. If the primary region fails and the team fails over, what is the approximate RPO?

Target word: recovery point objective **Decides:** Async replication loses roughly the lag.

A	Adjacent category	That is RTO.
B	Confusion set	Only synchronous replication approaches zero.
C	Correct	Writes not yet copied when the primary fails are lost.
D	Camouflage	The nightly backup does not set RPO when the replica is used.

8. Answer B Concept 5 • transfer

A business says it cannot lose any committed transactions if the primary fails. Which replication approach fits?

Target word: replication for RPO ≈ 0 **Decides:** No loss of committed data.

A	Inverted operation	Can lose the lag window.
B	Correct	Each commit is confirmed on the replica.
C	Dropped element	Could lose up to a day.
D	Adjacent category	Scale reads; usually asynchronous.

9. Answer B Concept 6

A marketing analytics database runs heavy queries 2 hours per business day, 22 business days a month. A provisioned instance costs \$2.00 an hour and runs all 730 hours of the month. A serverless option bills \$5.00 an hour, only while active. ~~The team gets alerts in a chat channel.~~ Which is cheaper, and what does it cost per month?

Target word: monthly cost comparison **Decides:** Active hours = 44.

A	Absolute language	Total cost depends on hours billed.
B	Correct	$44 \times \$5 = \220 vs. $730 \times \$2 = \$1,460$ provisioned.
C	Dropped element	Bills serverless for all 730 hours, ignoring pay-per-use.
D	Confusion set	Bills the provisioned instance only for active hours.

10. Answer C Concept 6 · transfer

A serverless database bills \$4 per active hour; a provisioned one costs a flat \$1,000 a month. Above how many active hours per month is provisioned cheaper?

Target word: break-even hours **Decides:** Flat cost / hourly rate.

A	Adjacent category	The hours in a month, not the break-even.
B	Wrong stem number	The flat cost read as hours.
C	Correct	$\$1,000 / \$4 = 250$.
D	Inverted operation	Multiplies instead of dividing.

11. Answer D Concept 7

At an e-commerce company ~~whose mobile app is rated 4.7 stars~~, analysts run queries that aggregate three years of order history across billions of rows, slowing the order-processing database during checkout. What architecture change fits best?

Target word: separate workloads **Decides:** Analytics is competing with transactions.

A	Absolute language	The business needs the analysis; it needs the right platform.
B	Confusion set	Key-value stores are poor at large aggregations.
C	Adjacent category	Slows writes and does not isolate the workloads.
D	Correct	Separates OLAP scans from OLTP traffic.

12. Answer A Concept 8

~~A 22-employee healthcare startup's managed cloud database, hosted in a SOC 2-audited region,~~ is found reachable from the public internet with a default admin account. Under the shared responsibility model, who must fix this, and how?

Target word: who owns the fix **Decides:** Access and configuration are the customer's responsibility.

A	Correct	Customers own configuration of access to their data.
B	Inverted operation	The provider secures infrastructure, not customer settings.
C	Absolute language	Default settings still need customer review.
D	Camouflage	The provider's audit does not cover the customer's configuration.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	2	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	9	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	8	
Absolute language	Always, never, guarantee — where judgment applies.	6	
Confusion set	Every option is real; the test is exact role-matching.	8	
Camouflage	Answers a stem fact that was never the question.	2	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Service models for databases	IaaS gives you a virtual machine you patch and manage; PaaS (managed database) handles patching, backups, and failover while you own schema and data; SaaS delivers a finished application.	Choosing a VM when the goal is to stop managing the database engine.	“Without managing patches or backups” but “design our own schema.”
2	Scaling reads	Read replicas offload read-heavy traffic from the primary; a high-availability standby is for failover, not usually for serving reads.	Adding write capacity or relying on a standby to serve reads.	A high read share and a busy primary.
3	Choosing a shard key	A good shard key has high cardinality and spreads load evenly; queries that do not include the key must hit every shard.	Sharding on a skewed or time-based key that creates hot partitions.	One shard far busier than the rest.
4	CAP trade-off	During a network partition a distributed system must choose consistency (refuse some requests) or availability (accept them and reconcile later).	Thinking a distributed system can drop partition tolerance or keep all three.	Regions losing contact plus a stated tolerance for stale or failed requests.
5	Replication and RPO	Asynchronous replication risks losing the replication lag worth of data ($RPO \approx \text{lag}$); synchronous replication gives RPO near zero at the cost of write latency. $RPO = \text{data loss}$; $RTO = \text{time to restore}$.	Confusing RPO with RTO, or assuming async replication loses nothing.	Replication mode and lag, or a “no lost transactions” requirement.
6	Pay-per-use vs. provisioned cost	Compare hours actually billed: provisioned bills every hour it runs; serverless bills only active time. Break-even hours = flat cost / hourly rate.	Comparing hourly rates instead of total monthly cost.	An intermittent workload with two pricing models.
7	OLTP vs. OLAP	Transactional databases serve many small reads and writes; large analytical scans belong in a columnar warehouse fed from the transactional system.	Running heavy analytics on the order-processing database.	Analytic queries slowing down transactions.
8	Shared responsibility for security	The cloud provider secures the infrastructure; the customer configures access, network exposure, credentials, and encryption settings for its data.	Assuming a managed service or provider certification makes the customer’s configuration secure.	A misconfiguration on a managed service.

Course scope (public description): Cloud Databases covers the application of cloud architectures to large-scale data systems. The course discusses the differences between cloud-native approaches to data architectures and smaller-scale systems. Students in this course apply cloud computing concepts to address specific business scenarios. There is no prerequisite for this course.